

# AIR. MADE VISIBLE.

Your ESP32 project, with four purposeful screens, a one-button interface and a corrected circuit you can build.

TAP TO EXPLORE. HOLD TO ACT.

04

OLED SCREENS

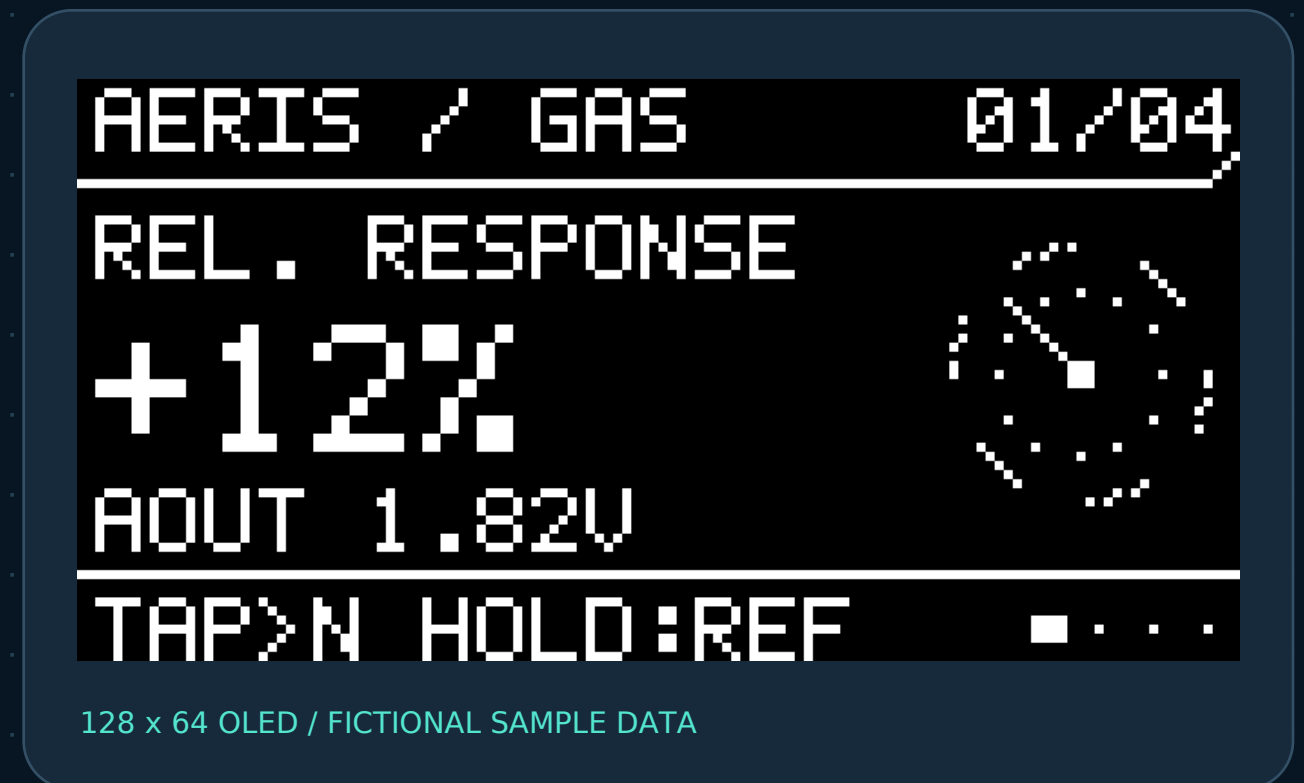
01

TACTILE BUTTON

02

SENSOR MODULES

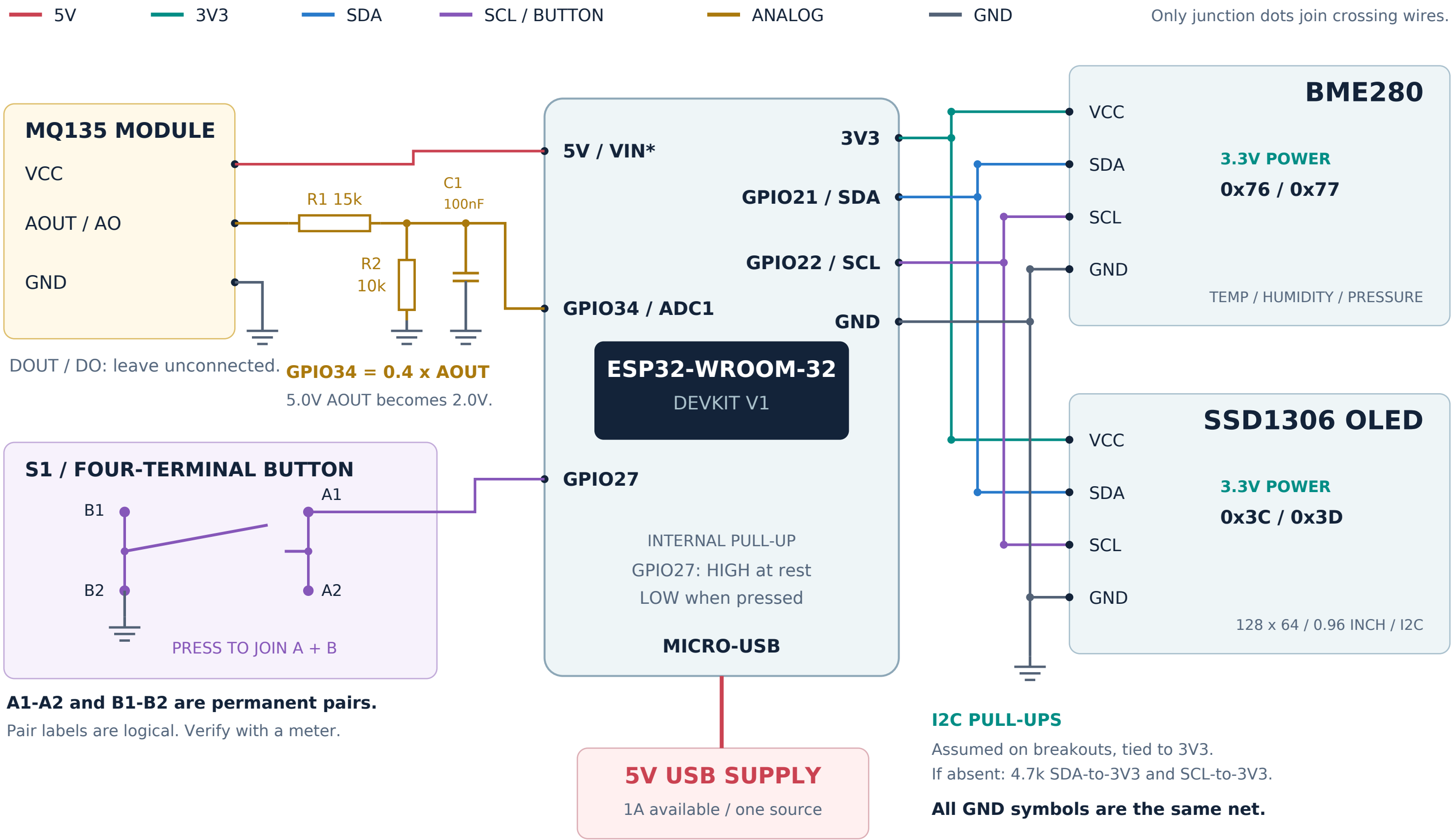
MQ135 gas response + BME280 climate. Package: Arduino sketch, PlatformIO profile, PDF + editable SVG circuit, shared-renderer previews and portable tests.



# AERIS / CIRCUIT

REV 1.0

ESP32 DevKit V1 + MQ135 + BME280 + SSD1306 + four-terminal push button



\*Verify USB 5V is available on your board pin and its current path is adequate. Never feed MQ135 AOUT directly to a GPIO.

Logical wiring schematic, not PCB layout. Check actual module labels. R1/R2: 1%, 1/4W. C1: near GPIO34. Power off before rewiring.

# Build the circuit with confidence.

Use the labels on your actual modules. GPIO numbers are not physical header positions.

## PARTS

| QTY    | ITEM                              |
|--------|-----------------------------------|
| 1      | ESP32-WROOM-32 / DevKit V1        |
| 1      | MQ135 module with analog output   |
| 1      | BME280 I2C breakout               |
| 1      | SSD1306 128x64 I2C OLED           |
| 1      | 4-terminal normally-open button   |
| 1 each | 15k + 10k resistors, 1%, 1/4W     |
| 1      | 100nF ceramic capacitor           |
| 1      | USB 5V supply, 1A available       |
| Set    | USB data cable, wires, breadboard |

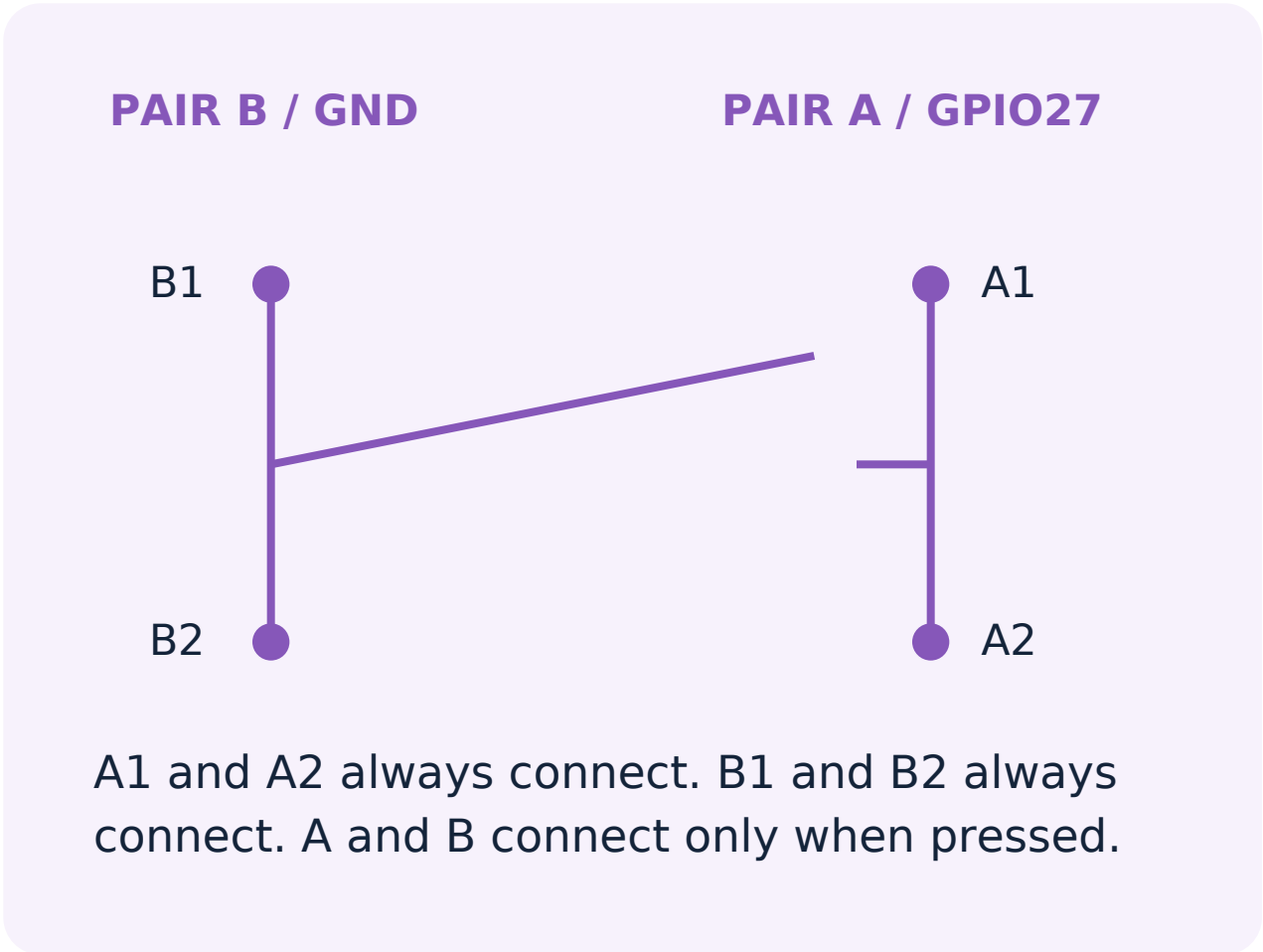
## EXACT CONNECTIONS

| MODULE / PIN           | CONNECT TO                 |
|------------------------|----------------------------|
| MQ135 VCC              | Verified USB 5V / VIN rail |
| MQ135 AOUT             | 15k -> GPIO34 node         |
| GPIO34 node            | 10k + 100nF to GND         |
| BME280 + OLED VCC      | ESP32 3V3                  |
| Both SDA pins          | GPIO21                     |
| Both SCL pins          | GPIO22                     |
| Button pair A / pair B | GPIO27 / GND               |
| All GND pins           | Common ground              |
| MQ135 DOUT / DO        | Leave unconnected          |

Before connecting power: check the 15k series / 10k shunt divider. The ADC node should be  $0.4 \times \text{AOUT}$ , approximately 2.0V at a 5.0V AOUT. Power the MQ135 from 5V and both I2C modules from 3V3. Keep the 100nF capacitor close to GPIO34.

# Four legs. Two electrical pairs.

The switch joins the two groups only while you press it. Pair labels below are logical.



## BUTTON ACTIONS

| GESTURE            | BEHAVIOR                         |
|--------------------|----------------------------------|
| Tap, then release  | Next screen; wraps after page 4  |
| Hold 2s on Gas     | Capture reference for 30 seconds |
| Hold 2s elsewhere  | Toggle normal / dim brightness   |
| Continue holding   | No repeated actions              |
| Release after hold | No additional page change        |

## FIND THE PAIRS WITH A MULTIMETER

With power off, use continuity mode. Each permanent pair beeps even without a press. Across the two groups it should beep only when pressed. Use one wire per group; duplicate legs need no extra wires.

Connect one A leg to GPIO27 and one B leg to GND. INPUT\_PULLUP makes the input HIGH at rest and LOW when pressed. Check the breadboard does not short the two groups. The navigation button is separate from BOOT/EN.

# Upload the firmware.

Arduino IDE is the simplest route. Keep all five files inside the AerisMonitor folder together.

## 01 Install the ESP32 board package

In Arduino IDE Preferences, add the official board URL linked below. Install esp32 by Espressif Systems in Boards Manager.

## 02 Install the Adafruit libraries

Use Library Manager for SSD1306, GFX Library, BME280 Library, Unified Sensor and BusIO. Accept their dependencies.

## 03 Open, select, compile

Open AerisMonitor/AerisMonitor.ino. Choose DOIT ESP32 DEVKIT V1 and the correct port. Click Verify, then Upload.

## 04 Read the startup report

Open Serial Monitor at 115200 baud. Look for the detected OLED and BME280 addresses. Press the new GPIO27 button to switch pages.

[Official ESP32 installation / board URL](#)

## BUILD PROFILE

**Target** Original ESP32 / WROOM-32

**Arduino core** 2.0.17 in pinned PIO profile

**PlatformIO** espressif32 6.9.0

**Board ID** esp32doit-devkit-v1

**Monitor** 115200 baud

**OLED** 0x3C, then 0x3D

**BME280** 0x76, then 0x77

Internet access is needed to install the real board package and libraries. This package contains source, not a prebuilt firmware binary.

## PLATFORMIO ALTERNATIVE

```
pio run --target upload
```

```
pio device monitor --baud 115200
```

# Four screens. One consistent HUD.

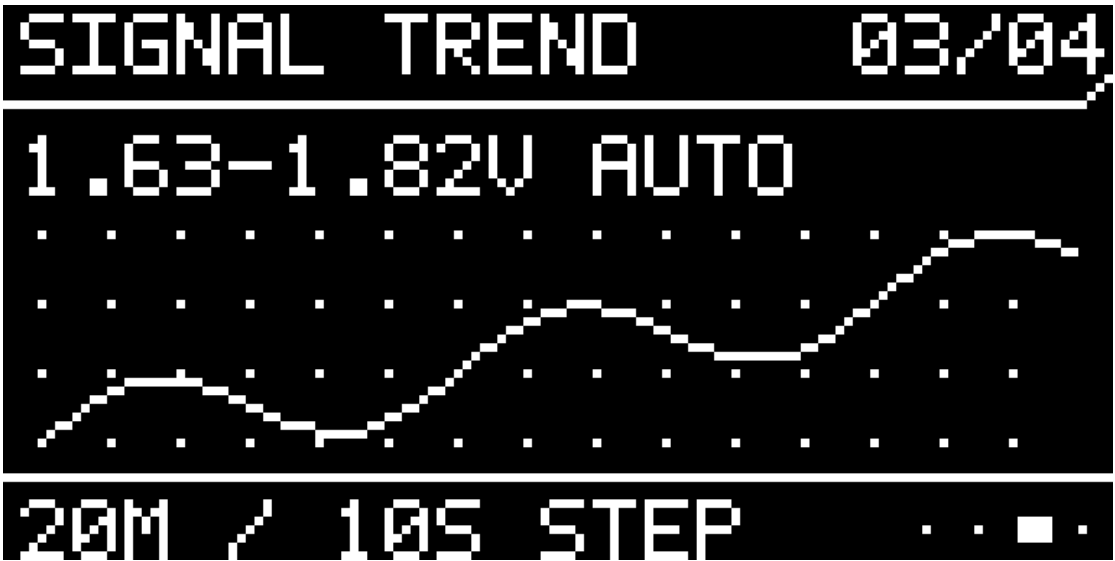
These are actual frames from Ui.h with fictional sample data. The physical OLED remains monochrome.

## 01 / GAS RESPONSE



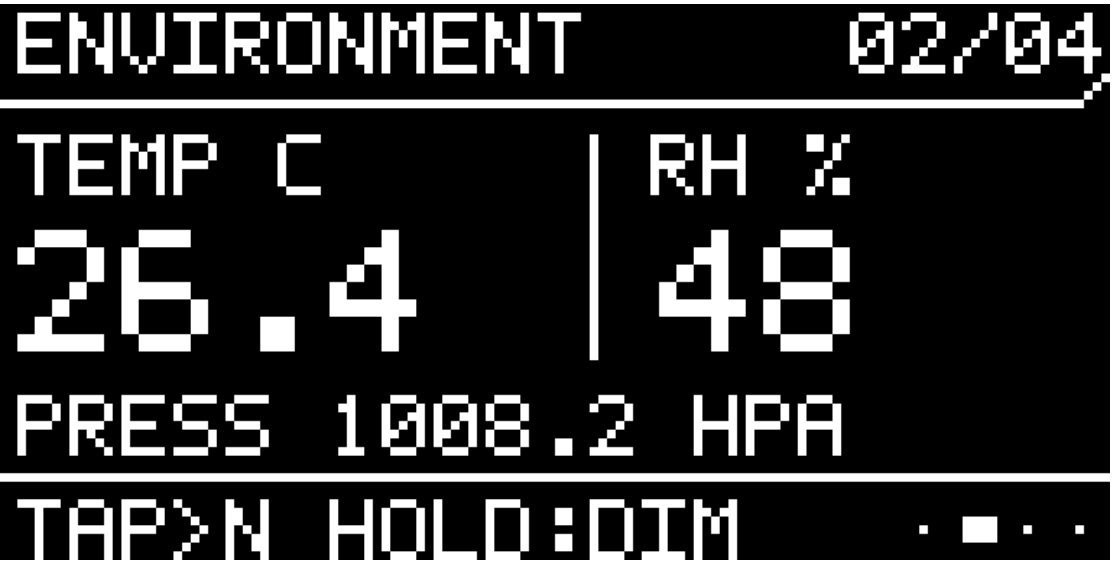
Signed change from your stored reference; raw AOUT and warm-up states.

## 03 / SIGNAL TREND



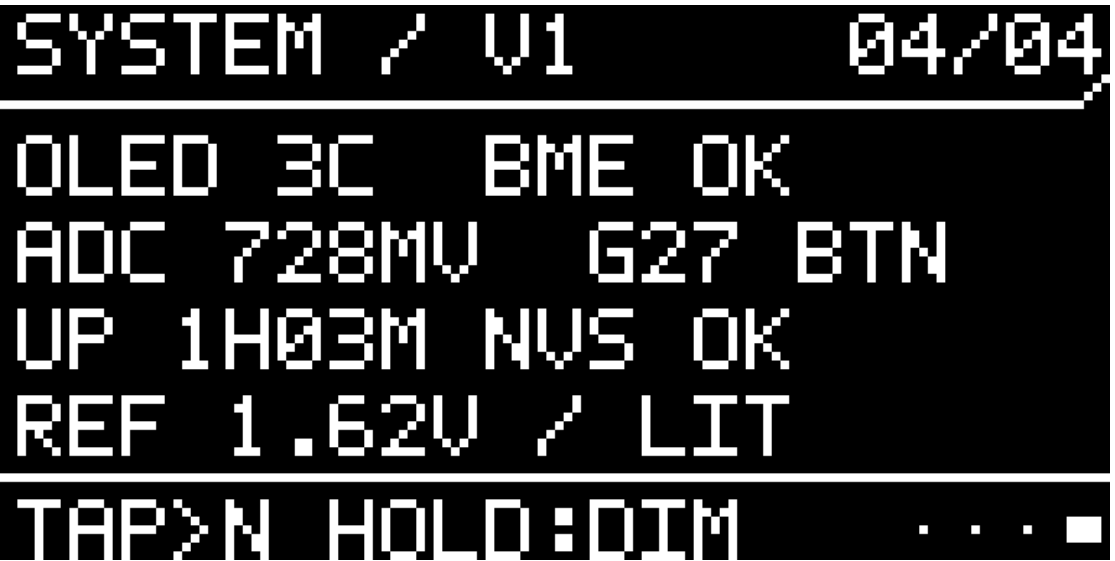
120 samples at 10-second intervals. Auto-scaled voltage, with gaps for missing data.

## 02 / CLIMATE



Temperature, relative humidity and local absolute pressure.

## 04 / SYSTEM STATUS



Detected display, BME status, ADC voltage, uptime and reference storage status.

# Let the sensor settle. Set a reference.

The MQ135 is a broad gas-response sensor; the firmware does not invent AQI or CO2 ppm.

## FIRST-USE SEQUENCE

- 01** Condition a fresh sensor. The default timer is 49 continuous hours, chosen to exceed the manufacturer's 48-hour initial preheat requirement. Longer storage may require longer aging; see README and the source manual.
- 02** Previously conditioned module? Set `MQ135_ALREADY_CONDITIONED` to true only when that exact sensor has completed conditioning. Later restarts use a 10-minute software settling gate; wait longer if it still drifts.
- 03** Capture clean ambient air. On the Gas page, hold for 2 seconds. Keep conditions stable for 30 seconds. A capture needs at least 80 samples and no more than 3% peak-to-peak voltage spread.
- 04** Read the result. `REFERENCE SAVED` means it survives power loss. An unstable capture keeps the old reference. `RAM ONLY` means saving failed and the new reference will be lost at restart.

## RELATIVE GAS RESPONSE

$$100 \times (V_{\text{now}} / V_{\text{ref}} - 1)$$

Example: 1.82V versus a 1.625V reference displays +12%. This is an electrical change, not 12% pollution or a health classification.

## WHAT THE HARDWARE CAN TELL YOU

The BME280 reports climate and local pressure. The MQ135 reports a nonselective response affected by gases, humidity, temperature, warm-up and drift. No gas identification or unvalidated climate compensation is applied.

The heater timer assumes continuous shared power; it cannot measure heater current. When replacing the MQ135, erase old Preferences/flash before reconditioning and capturing a new reference.

Educational monitor. Do not use it as a certified smoke/gas alarm or test it with flames or lighter-gas spray.

# Bring-up checks and troubleshooting.

Power down before moving wires. Then use the Serial report and the circuit sheet to isolate the fault.

| SYMPTOM                | CHECK                                                                                 |
|------------------------|---------------------------------------------------------------------------------------|
| Blank OLED             | 3V3/GND, SDA21/SCL22, 0x3C or 0x3D; confirm SSD1306, not SH1106.                      |
| Button does nothing    | One wire from each electrical pair; GPIO27, not a numbered header position.           |
| BME280 unavailable     | Genuine BME280, 0x76/0x77, I2C straps and power. BMP280 cannot read humidity.         |
| ADC out of range       | MQ135 5V, common ground, 15k/10k orientation, 100nF and GPIO34 voltage.               |
| Resets with heater on  | USB supply capacity, cable voltage drop and the board's USB-to-VIN current path.      |
| Warm-up / no reference | Allow conditioning and settling; then hold on Gas. Do not interpret raw volts as AQI. |
| Unstable reference     | Wait for less drift; keep ambient conditions steady. The old reference is retained.   |

## VERIFIED HERE

18 portable C++ behavioral/render checks passed. Full sketch syntax checked using desktop API stubs. Preview pixels and PDF layouts reviewed. ESP32 target compilation, upload and physical sensor testing remain to be done.

See README.md for setup commands and serial controls. Bench acceptance checks are in docs/VALIDATION.md.

## PRIMARY REFERENCES / CLICK TO OPEN

- [Winsen MQ135 electrical data and conditioning](#)
- [Espressif ADC and calibrated millivolt API](#)
- [Espressif GPIO and internal pull-ups](#)
- [Adafruit BME280 pinout and I2C addresses](#)
- [PlatformIO ESP32 DevKit V1 board profile](#)